Galaxy Telecom IT Service Desk Agent

COPILOT STUDIO · AZURE FUNCTIONS · AZURE SQL · MICROSOFT GRAPH · MICROSOFT TEAMS

An AI-powered IT support agent for Galaxy Telecom employees, deployed to Microsoft Teams. Employees describe IT issues in natural language, receive guided troubleshooting from a SharePoint knowledge base, and raise support tickets through conversation. Built on Microsoft Copilot Studio with a FastAPI backend on Azure Functions and Entra-only authentication throughout — no passwords stored anywhere.

Microsoft Copilot Studio | Azure Functions | Azure SQL | Microsoft Graph API | Microsoft Teams | FastAPI | Python 3.12 | Managed Identity

Gayatri Anne

AI & Cloud Architect | 18+ Years Enterprise IT

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The Problem

IT helpdesk teams in enterprise telecoms are overwhelmed with repetitive support requests. Employees waste time raising tickets through clunky portals, waiting for acknowledgement emails, and chasing status updates — while IT staff manually triage and assign work.

The Challenge

Employees have no fast, conversational way to report IT issues. Existing portals require navigating forms, selecting categories, and describing problems in structured fields — a process that is slow, error-prone, and frustrating. Status updates require logging back in.

The Solution

A single Teams-based AI assistant that handles the full support lifecycle through natural language. Employees describe their issue; the agent troubleshoots first using the IT knowledge base. If unresolved, it raises a ticket, sends a confirmation email, and lets the employee check status — all in the same chat window.

3	0	3	Teams
API ENDPOINTS	PASSWORDS STORED	COPILOT TOOLS	DEPLOYMENT CHANNEL

KEY FEATURES

- Natural language troubleshooting
- Zero-form ticket raising
- Instant status lookup
- Managed Identity auth
- Confirmation email via Graph API
- SharePoint knowledge base

Step by Step

One agent, three capabilities. Copilot Studio routes automatically based on what the employee asks.

01

EMPLOYEE SENDS A MESSAGE IN TEAMS

The employee types a natural language description of their IT issue — for example, "My VPN is not connecting" or "My laptop won't start."

02

AGENT TROUBLESHOOTS USING SHAREPOINT KNOWLEDGE BASE

Copilot Studio searches the IT-Knowledge-Base SharePoint site for relevant troubleshooting guidance and presents step-by-step resolution steps to the employee.

03

EMPLOYEE CONFIRMS ISSUE IS UNRESOLVED

If steps fail, the employee says so. The agent confirms intent to raise a ticket and collects all required details from the conversation context.

04

FASTAPI INSERTS TICKET INTO AZURE SQL

The agent calls the RaiseTicket tool via the Power Platform Custom Connector. The FastAPI endpoint on Azure Functions inserts the ticket row. Azure SQL generates the ticket number automatically using a computed column (INC-XXXX format).

05

MICROSOFT GRAPH API SENDS CONFIRMATION EMAIL

Immediately after insert, FastAPI calls Microsoft Graph /sendMail using a token obtained from Managed Identity. The employee receives a formatted HTML email with full ticket details — no Power Automate, no passwords.

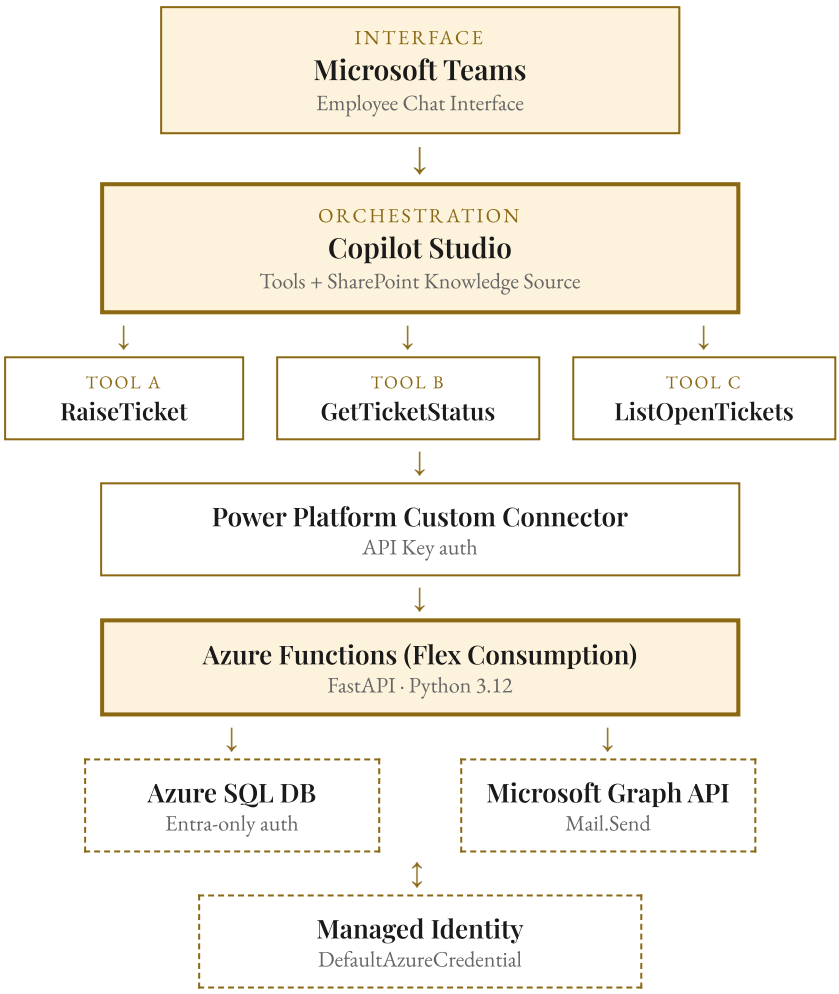
06

EMPLOYEE CAN CHECK STATUS OR LIST OPEN TICKETS

At any point the employee can ask "show me my open tickets" or "what is the status of INC-0014" — the agent calls the appropriate tool and returns live data from Azure SQL, filtered to that employee's own tickets only.

System Design

Built on Azure with Entra-only authentication throughout. No SQL passwords. No hardcoded keys. Managed Identity handles all service-to-service auth.



Ticket Creation

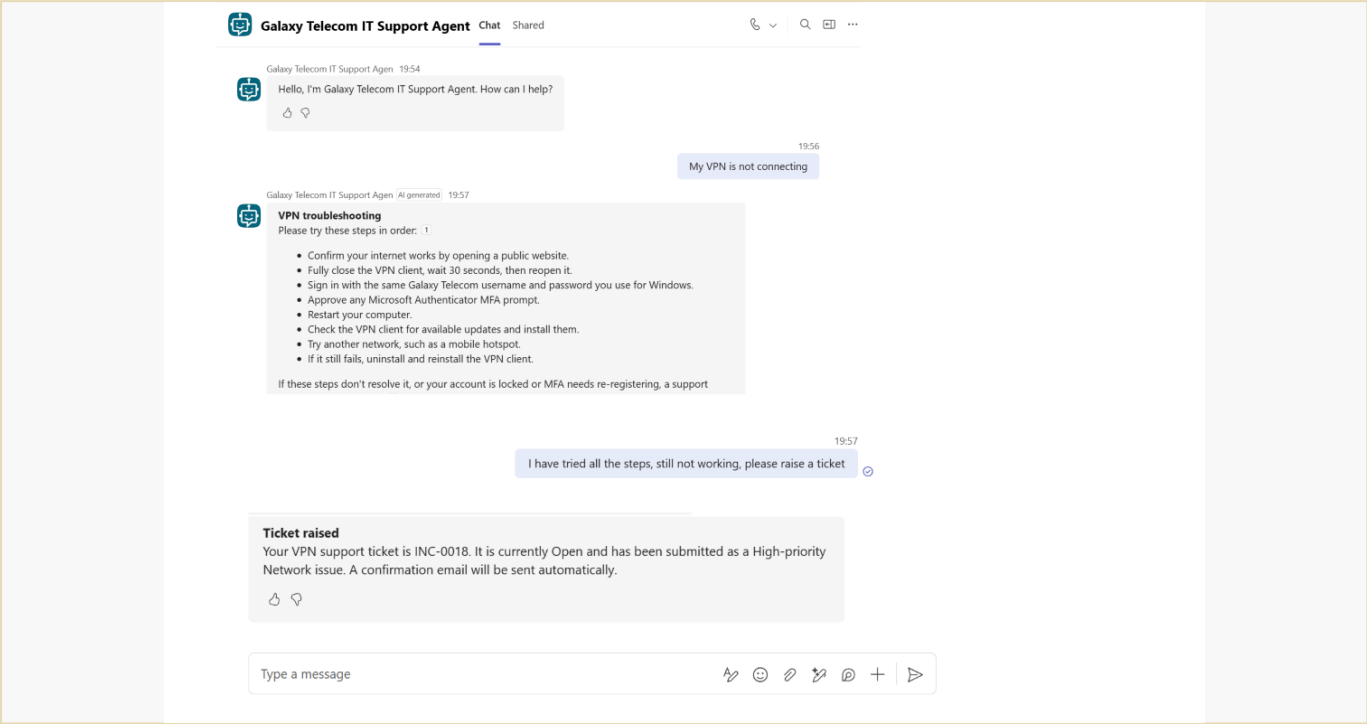
The Custom Connector calls POST /tickets. FastAPI inserts the row via pyodbc using a token from DefaultAzureCredential. Azure SQL returns ticket_id and the computed ticket_number. FastAPI immediately calls Graph /sendMail — email failure never blocks ticket creation.

Status and Listing

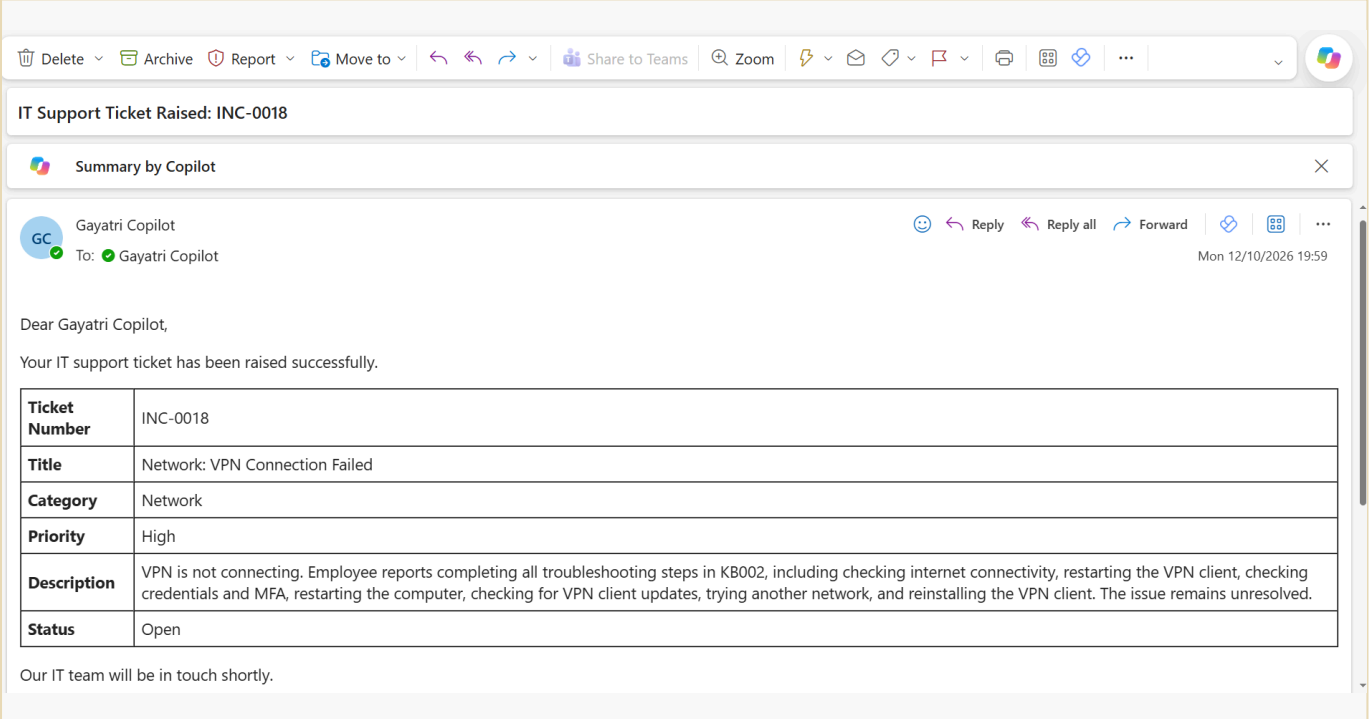
GET /tickets/{ticket_number} filters by both ticket_number AND employee_email — an employee cannot look up another employee's ticket. employee_email is injected automatically from System.User.Email; the employee never types it.

Agent in Teams

All three capabilities tested end-to-end in Microsoft Teams with live Azure SQL data. Responses are consistent, accurate and arrive in seconds.

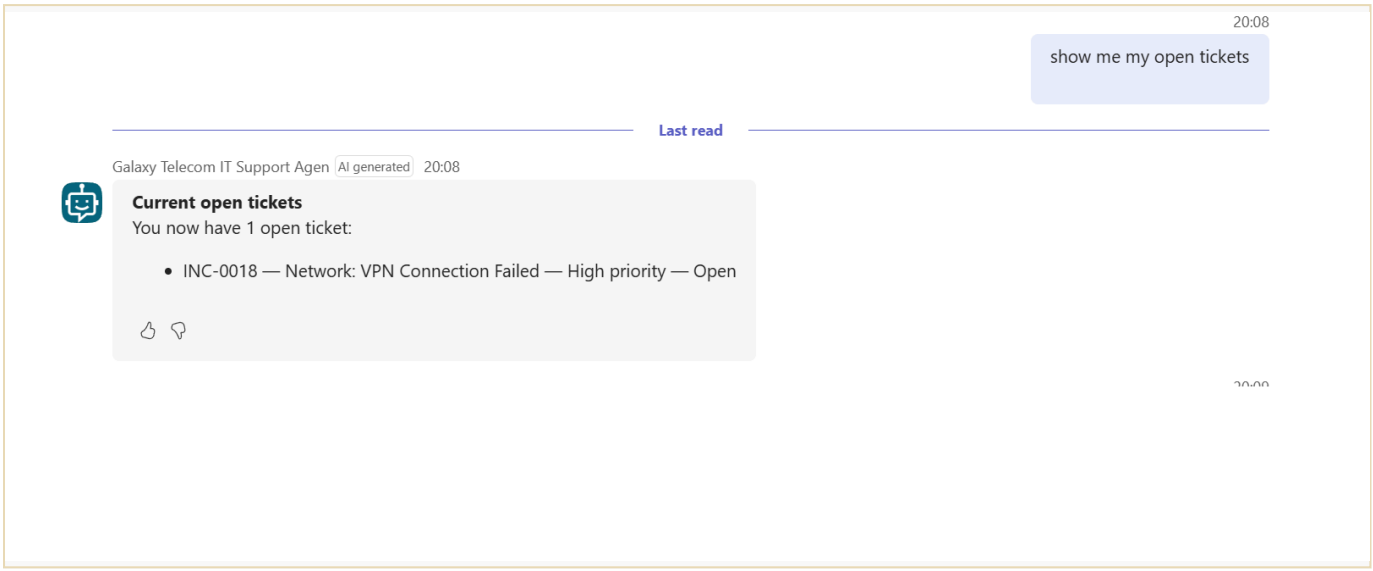


Employee reports VPN issue, agent troubleshoots from SharePoint KB, then raises INC-0018 after steps fail

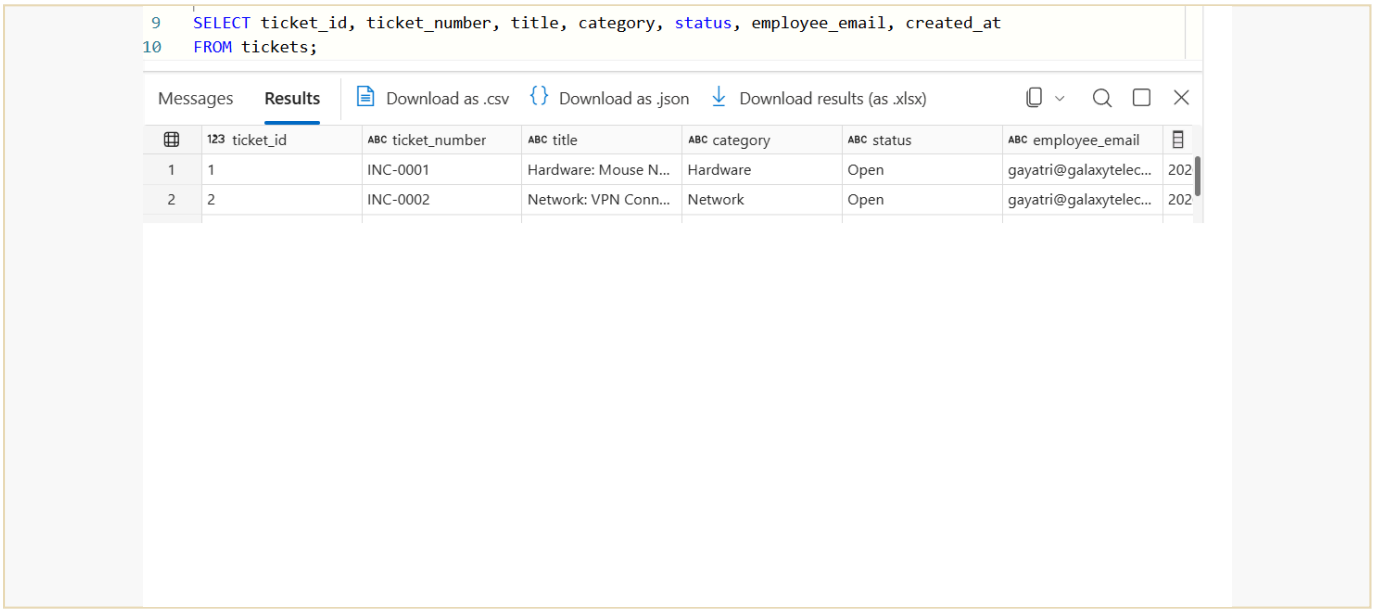


HTML confirmation email delivered via Microsoft Graph API — ticket number, category, priority and description all correctly populated from the SQL insert

Agent & Azure Views



Employee asks for open tickets — agent returns live results from Azure SQL filtered to their email



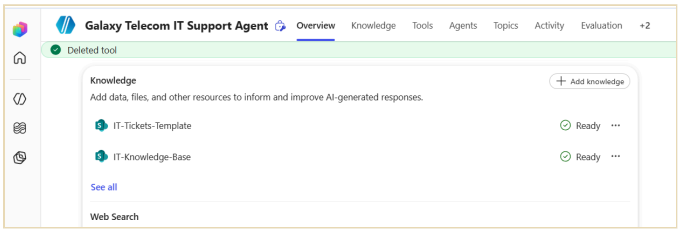
Azure SQL tickets table — ticket_number computed column auto-generates INC-XXXX from IDENTITY sequence



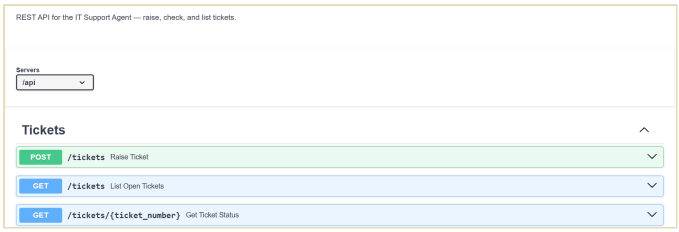
Employee queries status of INC-0006 — agent returns title, priority and status directly from Azure SQL

Components

COMPONENT	TECHNOLOGY	ROLE
AI Agent	Microsoft Copilot Studio	Hosts the assistant, manages tools, SharePoint knowledge source and system prompt
API Layer	FastAPI on Azure Functions	Exposes three REST endpoints for ticket operations; FUNCTION auth level
Database	Azure SQL (Entra-only)	Stores all ticket data; ticket_number generated by computed column
Authentication	Microsoft Entra ID	Entra-only SQL auth; DefaultAzureCredential for all service tokens
Email	Microsoft Graph API	Sends HTML confirmation emails using Managed Identity token — no Power Automate
Connector	Power Platform Custom Connector	Bridges Copilot Studio tools to Azure Functions API with API Key security
Deployment	Azure Functions Flex Consumption	Serverless, Linux, Python 3.12; scales to zero when idle
Knowledge Source	SharePoint (IT-Knowledge-Base)	Provides troubleshooting content for the agent's generative answers



Copilot Studio — Knowledge & Tools overview



FastAPI Swagger UI — POST /tickets, GET /tickets, GET /tickets/{ticket_number}

Key Design Decisions

- Managed Identity for all Azure auth — DefaultAzureCredential works locally and on Azure with zero config change
- Email via Graph API not Power Automate — eliminates field mapping ambiguity; values come directly from the SQL response
- employee_email injected from System.User.Email — employees never type their own email; prevents cross-employee lookup
- Entra-only SQL — no SQL admin account exists; access granted via CREATE USER FROM EXTERNAL PROVIDER

Future Enhancements

- Automatic ticket assignment to IT staff based on category and priority
- Ticket update notifications — employee notified in Teams when status changes
- Extend to support asset requests (laptop, monitor, phone)
- SLA breach alerts when high-priority tickets remain open beyond threshold
- Power BI dashboard for IT manager ticket analytics

Gayatri Anne

AI & Cloud Architect | 18+ Years Enterprise IT

I build AI-powered automation systems that eliminate manual work from business processes, combining agentic workflows, large language models and cloud integration to deliver production-ready solutions.

CERTIFICATIONS

- Azure Solutions Architect Expert
- Azure AI Engineer Associate
- Azure Administrator
- Azure AI Fundamentals
- Azure Fundamentals
- TOGAF Foundation
- Python (PCAP)
- Microsoft Certified: AI Agent Builder Associate